Brainstorming Stochastic Differential Equations

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Disclaimer: The equations presented here are taken directly from the original paper. Some derivations in the paper contain errors, which are discussed in detail in document D-A17212-3016. This outline is intended to serve as a structural summary of the paper's framework, which we will use as a reference as we brainstorm and develop our own model.

D-A17212-3016's SDEs

Linear Risk Free Asset

$$dP_0(t) = (\alpha + \sigma t)P_0(t)\,dt, \tag{1}$$

- $P_0(t)$: The price (or value) of an asset at time t.
- $dP_0(t)$: The infinitesimal change in $P_0(t)$ over the time interval $[t, t + dt]$.
- α: A constant representing the baseline (deterministic) growth rate of the asset.
- σ: A constant representing the rate at which the growth accelerates over time.
- t: Time, typically measured continuously from $t = 0$.
- $(\alpha + \sigma t)$: The time-dependent growth rate of the asset — it increases linearly over time.
- $(\alpha + \sigma t)P_0(t)\,dt$: The deterministic change in $P_0(t)$ over a small time interval dt, proportional to both the current price and the time-varying growth rate.

Understanding the Role of dt

Suppose you're driving at a speed of 60 mph. Then:

- Your rate of change of position is 60 miles per hour.
- If you drive for a tiny amount of time, say 0.01 hours (36 seconds), the distance you travel is:

$$\text{distance} = \text{rate} \times \text{time} = 60 \times 0.01 = 0.6 \text{ miles}$$

Likewise, in the differential equation:

$$dP_0(t) = \text{rate} \times dt = (\alpha + \sigma t)P_0(t)\,dt, \tag{1}$$

the differential dt converts the instantaneous rate into an actual change. It represents an infinitesimal increment in time and serves as the "unit of time" required to accumulate the effect of the rate.

Ornstein-Uhlenbeck Risk Free Asset

$$dr_t = \alpha(\beta - r_t)\,dt + \sigma\,dZ_t, \quad r_{t_0} = r_0 \tag{31a}$$

- dr_t: The infinitesimal change in the short rate r_t over a small time interval dt; this represents the total stochastic differential of the process.
- $\alpha(\beta - r_t)\,dt$: The drift term, representing deterministic mean-reverting behavior.
- α: Speed of mean reversion — how quickly r_t is pulled back toward the long-run mean.
- β: Long-run average level of the short rate.
- $(\beta - r_t)$: The distance from the current rate to its long-run mean.
- $\sigma\,dZ_t$: The diffusion term, which introduces randomness to the evolution of r_t.
- σ: Volatility coefficient — scales the magnitude of the random shocks.
- dZ_t: Increment of a standard Brownian motion (Wiener process), capturing the stochastic noise.
- $r_{t_0} = r_0$: Initial condition specifying that the short rate at initial time t_0 is equal to the given constant r_0.

Risky Asset

$$dP_1(t) = P_1(t)\left(\mu\,dt + \sigma\,dZ(t)\right), \tag{2}$$

- $P_1(t)$: The price of the **risky asset** at time t.
- $dP_1(t)$: The infinitesimal change in the risky asset's price over a small time increment dt.
- $\mu\,dt$: The deterministic drift term, representing the expected rate of return of the risky asset.
- $\sigma\,dZ(t)$: The stochastic (random) component of price changes. Here, σ is the volatility of the asset, and $dZ(t)$ is an increment of standard Brownian motion, normally distributed with mean zero and variance dt.

The Wealth Process

$$dW^S(t) = S(t)\frac{dP_1(t)}{P_1(t)} + (W(t) - S(t))\,\frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)\,S(t)\,dt. \tag{3}$$

- $S(t)\dfrac{dP_1(t)}{P_1(t)}$: This term represents the return earned on the portion $S(t)$ of wealth invested in the risky asset. It captures both the deterministic drift and the stochastic fluctuation from the risky asset price process.
- $(W(t) - S(t))\,\dfrac{dP_0(t)}{P_0(t)}$: This is the return on the remaining wealth invested in the risk-free asset. Since the risk-free asset evolves deterministically, this term contributes only to the drift of the wealth process.
- $-(\vartheta + \beta - d)S(t)dt$: This term adjusts for the costs and income associated with holding the risky asset. It includes transaction costs ϑ, taxes β, and dividend income d. The net effect is a reduction in drift due to market frictions.

After substituting the Linear Risk Free Asset Equation and the Risky Asset Equation, we have:

$$dW^S(t) = \{W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t) - \vartheta - \beta]\,S(t)\}\,dt + \sigma S(t)\,dZ(t) \tag{4}$$

Section Three Framework

Let $f(t, X_t) = V(t, W(t)) = \mathbb{E}^{t,w}\left[U(W^S(T))\right]$ be the value function.

Applying Ito's lemma to the value equation gives

$$\begin{aligned} dV(t, W(t)) = \Bigg[&\frac{\partial V}{\partial t} + \frac{\partial V}{\partial w} \cdot (W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t) - \vartheta - \beta]\, S(t)) \\ &+ \frac{1}{2} \cdot \frac{\partial^2 V}{\partial w^2} \cdot \sigma^2 S^2(t) \Bigg] dt + \sigma S(t) \cdot \frac{\partial V}{\partial w} \cdot dZ(t). \end{aligned}$$

Applying expectation to $dV(t, W(t))$ and setting the result to zero gives us the HJB:

$$J_t + J_w \left\{[W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t + \vartheta + \beta)]S(t)]\right\} + J_{ww} \left[\frac{S^2(t)\sigma^2}{2}\right] = 0. \tag{9}$$

Differentiating the HJB with respect to S(t) gives

$$S^*(t) = \frac{[(\vartheta + \beta + \alpha + \sigma t) - (\mu + d)]J_w}{\sigma^2 J_{ww}}. \tag{12}$$

Eliminating Dependency on W

Using the bolded condition,

$$J(t, s, w) = \max_S \left\{\mathcal{B}^S V(t, w)\right\} = 0; \quad \mathbf{V}(\mathbf{T}, \mathbf{w}) = \mathbf{U}(\mathbf{w}), \quad 0 < t < T, \tag{8}$$

we propose

$$J(t, s, w) = h(t, T)\, \frac{w^{1-\phi} - k}{1 - \phi}. \tag{13}$$

Therefore,

$$J_t = h_t \frac{w^{1-\phi} - k}{1 - \phi}, \quad J_w = h w^{-\phi}, \quad \text{and} \quad J_{ww} = -\phi w^{-\phi-1}. \tag{15}$$

The HJB equation becomes

$$\begin{aligned} &h_t \frac{w^{1-\phi} - k}{1 - \phi} + h w^{-\phi} \left\{W(t)(\alpha + \sigma t) + [(\mu + d) - (\alpha + \sigma t + \vartheta + \beta)]\, S^*(t)\right\} \\ &+ \sigma S^*(t) dZ(t) - \phi w^{-\phi-1} h \left[\frac{S^{*2}(t)\sigma^2}{2}\right] = 0. \end{aligned} \tag{16}$$

Differentiating with respect to S(t) gives

$$S^*_{d,\vartheta,\beta}(t) = \frac{[(\mu + d) - (\alpha + \sigma t + \vartheta + \beta)]}{\phi\sigma^2} w. \tag{18}$$

Section Four Framework

Start with the redefined stochastic differential equation (SDE) for wealth

$$\begin{aligned} dW^S(t) &= S(t) \cdot \frac{dP_1(t)}{P_1(t)} + (W(t) - S(t)) \cdot \frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t)\, dt \\ &= S(t)\, (\mu\, dt + \sigma\, dZ_t) + (W(t) - S(t))\, r(t)\, dt - (\vartheta + \beta - d)S(t)\, dt. \end{aligned}$$

Let the value function be $v(t, r(t), W(t))$ and apply multivariable Itô's Lemma to derive dv.

Take expectations and apply the Bellman Principle to obtain the Hamilton–Jacobi–Bellman (HJB) equation:

$$\begin{aligned} J_t + \alpha(\beta - r_t)J_r + [W_t + (\mu + d) - (r_t + \sigma + \beta)]S(t)]J_w + r_t^2 J_{rw} \\ + \frac{1}{2}J_{rr} + \frac{1}{2}\sigma^2 S(t)^2 J_{ww} = 0. \end{aligned} \tag{32}$$

Differentiate the HJB with respect to the control S to derive the first-order condition

$$[(\mu + d) - (r_t + \vartheta + \beta)]\, J_w + \sigma^2 S(t) J_{ww} = 0. \tag{33}$$

Solve for S^*, the initial form of the optimal policy

$$S^* = \frac{[(\mu + d) - (r_t + \vartheta + \beta)]J_w}{\sigma^2 J_{ww}} - \frac{\sigma^2 J_{rw}}{\sigma^2 J_{ww}}. \tag{34}$$

Eliminate the dependency on wealth w by conjecturing

$$J(t, r, w) = G(t, r) \cdot \frac{w^{1-\phi}}{1-\phi}, \quad \phi \neq 1. \tag{35a}$$

Use the following partial derivatives (from Eq. 35c)

$$J_w = w^{-\phi} G, \quad J_{ww} = -\phi w^{-\phi-1} G, \quad J_{rw} = w^{-\phi} G_r. \tag{35c}$$

Plug these expressions into the first-order condition to obtain

$$S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu + d) - (r_t + \vartheta + \beta)]}{\sigma^2} + \frac{G_r}{G}\right]. \tag{36}$$

Remove the dependency on r by letting

$$G(t, r) = h(t)\left(\frac{r^{1-\phi}}{1-\phi}\right), \quad h(T) = \frac{1-\phi}{r^{1-\phi}}. \tag{37a, 37b}$$

Using $G_r = r^{-\phi} h(t)$ and simplifying

$$\frac{G_r}{G} = \frac{1-\phi}{r} = (1-\phi) r_t.$$

Final form of the optimal control

$$S^*_{d,\vartheta,\beta}(t) = \frac{w}{\phi}\left[\frac{[(\mu + d) - (r_t + \vartheta + \beta)]}{\sigma^2} + (1-\phi) r_t\right]. \tag{38}$$

James' Proposed Equations

Risk Free Asset

Cox–Ingersoll–Ross (CIR) Model

$$dr_t = \kappa(\theta - r_t)\, dt + \sigma\sqrt{r_t}\, dZ_t$$

- r_t: The short-term interest rate at time t.
- κ: Speed of mean reversion — how quickly r_t moves toward the long-run mean.
- θ: Long-run mean level of the short rate.
- σ: Volatility parameter; higher σ means more random fluctuations.
- $\sqrt{r_t}$: State-dependent volatility; ensures non-negative rates.
- dZ_t: Brownian motion capturing randomness.

Key Features of the CIR Model

Feature	Description
Mean-Reversion	Yes. Drift term $\kappa(\theta - r_t)$ pulls rates toward the long-run mean θ.
Volatility	State-dependent: volatility increases with $\sqrt{r_t}$.
Positivity	The process is always non-negative. It remains strictly positive if the Feller condition $2\kappa\theta \geq \sigma^2$ holds. Otherwise, zero can be reached but is reflecting.
Bond Pricing Tractability	Closed-form solutions exist under certain assumptions, useful for analytical pricing.
Economic Realism	More realistic than OU for modeling interest rates due to non-negativity and proportional volatility.

Risk-Free Rate: Asymmetric Cyclical Drift Model

$$dr_t = \alpha \left|\sin(\omega t)\right| \, dt + \bar{r} \, dt + \sigma \, dZ_t$$

- r_t: Short-term interest rate at time t.
- α: Amplitude of the cyclic drift spikes.
- ω: Frequency of the cycle (e.g., business cycle periodicity).
- $\bar{r}$: Long-run baseline level of the interest rate.
- σ: Volatility of the rate due to market shocks.
- dZ_t: Standard Brownian motion.

Intuition

This model improves realism by generating asymmetric cycles: interest rates stay near $\bar{r}$ for extended periods and experience brief upward surges due to the $|\sin(\omega t)|$ term. It reflects empirical behavior of monetary policy and bond yields — which often remain low or stable for long durations and spike briefly in response to inflation or central bank shocks.

Key Features of the Asymmetric Cyclical Model

Feature	Description
Asymmetric Drift	The drift term $\alpha|\sin(\omega t)|$ produces **sharp rate spikes** and **longer low-rate plateaus**.
Positive Baseline	The term $\bar{r}$ ensures the rate oscillates around a **realistic long-run mean**, avoiding zero as the center.
Non-Mean-Reverting	The model does not pull r_t toward a fixed equilibrium but rather cycles upward and returns naturally.
Exogenous Cycles	Cycle timing is externally imposed via ω, enabling control over frequency of rate shocks.
Constant Volatility	The stochastic term σdZ_t injects randomness independent of current rate levels.

Risky Asset

Stochastic Drift and Volatility Model

$$dP_1(t) = P_1(t)\,[\mu_t\,dt + \sigma_t\,dZ_t]$$

where both the **drift** μ_t and the **volatility** σ_t evolve according to their own stochastic differential equations:

$$d\mu_t = \kappa_\mu(\theta_\mu - \mu_t)\,dt + \xi_\mu\,dW_t^{(1)}$$
$$d\sigma_t = \kappa_\sigma(\theta_\sigma - \sigma_t)\,dt + \xi_\sigma\,dW_t^{(2)}$$

- $P_1(t)$: Price of the risky asset at time t.
- μ_t: Instantaneous expected return (drift), governed by a mean-reverting process.
- σ_t: Instantaneous volatility, also governed by a mean-reverting process.
- dZ_t: Standard Brownian motion driving asset price shocks.
- $dW_t^{(1)}, dW_t^{(2)}$: Independent Brownian motions driving shocks to μ_t and σ_t, respectively.
- $\kappa_\mu, \kappa_\sigma$: Speeds of mean reversion for μ_t and σ_t.
- $\theta_\mu, \theta_\sigma$: Long-run mean levels of drift and volatility.
- ξ_μ, ξ_σ: Volatilities of the drift and volatility processes.

Key Features

Feature	**Description**
Stochastic Drift	The drift μ_t fluctuates around a long-term mean, capturing cyclical excess returns or macroeconomic shocks.
Stochastic Volatility	The volatility σ_t evolves dynamically, modeling realistic volatility clustering.
Mean-Reversion	Both μ_t and σ_t revert to their respective long-run means.
Multiple Shocks	Separate Brownian motions for drift and volatility allow for rich, multidimensional randomness.
Financial Realism	Suitable for capturing momentum, regime-switching, or liquidity-driven effects.

Risky Asset: Jump-Diffusion Model (Merton Type)

$$dP_1(t) = P_1(t^-)\,[\mu\,dt + \sigma\,dZ_t + (J-1)\,dN_t]$$

Term Definitions

- $P_1(t)$: Price of the risky asset at time t.
- $P_1(t^-)$: Left-limit of P_1 just before a potential jump.
- $\mu\,dt$: Expected rate of return (deterministic drift).
- $\sigma\,dZ_t$: Continuous random fluctuations modeled by Brownian motion.

- dN_t: Poisson jump process with intensity λ; equals 1 if a jump occurs during $[t, t+dt)$, otherwise 0.
- J: Random variable representing the size of the jump (e.g., lognormally distributed).
- $(J-1)dN_t$: Represents a multiplicative jump. If $J > 1$, it's a jump up; if $J < 1$, it's a jump down.

Key Features of the Jump-Diffusion Model

Feature	**Description**
Discontinuous Shocks	The Poisson process introduces discrete, sudden jumps in asset price.
Realistic Dynamics	More accurate for modeling equities, cryptocurrencies, or commodities exposed to sudden macro shocks.
Drift Compensation	The expected jump size and intensity affect the overall drift; often adjusted as $\mu' = \mu - \lambda \mathbb{E}[J-1]$ to preserve risk-neutral valuation.
Parameterization Flexibility	Jump frequency λ, jump size distribution J, and continuous vol σ are all tunable for different asset classes.
Closed-form Pricing (under assumptions)	For some derivatives (e.g., European options), closed-form solutions still exist (Merton 1976).

Wealth Process

Wealth Process with Consumption Term

We modify the existing wealth process to account for continuous consumption $C(t)$, which reduces the investor's wealth over time.

$$
\begin{aligned}
dW^S(t) &= S(t) \cdot \frac{dP_1(t)}{P_1(t)} + (W(t) - S(t)) \cdot \frac{dP_0(t)}{P_0(t)} - (\vartheta + \beta - d)S(t)\,dt - C(t)\,dt \\
&= S(t)\,(\mu\,dt + \sigma\,dZ_t) + (W(t) - S(t))\,r(t)\,dt - (\vartheta + \beta - d)S(t)\,dt - C(t)\,dt.
\end{aligned}
$$

Term Breakdown

- $S(t)\frac{dP_1(t)}{P_1(t)}$: Return from wealth invested in the risky asset.
- $(W(t) - S(t))\frac{dP_0(t)}{P_0(t)}$: Return from the wealth invested in the risk-free asset.
- $-(\vartheta + \beta - d)S(t)\,dt$: Adjustment for transaction costs, taxes, and dividends.
- $-C(t)\,dt$: **New term** representing the rate of consumption, which directly reduces wealth over time.
- $C(t)$: Control variable representing the investor's consumption rate at time t.

Key Features of Wealth SDE with Consumption

Feature	**Description**
Investment Allocation	Splits wealth between risky and risk-free assets via control $S(t)$.
Consumption Modeling	Incorporates realistic behavior such as spending on goods and services via $C(t)$.
Wealth Depletion	Consumption reduces future wealth; overconsumption can lead to insolvency.
Intertemporal Tradeoff	Investors must balance current utility from consumption with future investment gains.
Control Problem	Optimal control now includes both $S(t)$ and $C(t)$, increasing the dimension of the HJB.

Modeling the Consumption Term $C(t)$

The consumption term $C(t)$ represents the investor's outflow toward consumption over time and plays a central role in shaping the intertemporal tradeoff between current utility and future wealth. Several modeling choices are available depending on the economic context and desired tractability:

Deterministic Consumption Path

Specify $C(t)$ as a deterministic function of time:

- **Constant consumption:** $C(t) = c$
- **Linearly increasing consumption:** $C(t) = c_0 + c_1 t$
- **Cyclical consumption:** $C(t) = A \sin(\omega t + \phi)$

Proportional to Wealth

Assume consumption is a fixed proportion of wealth:

$$C(t) = \kappa W(t), \quad \kappa \in (0, 1)$$

Interpretation: The investor consumes a constant fraction of their wealth over time.
Advantage: This model aligns well with empirical consumption behavior.

Endogenous (Optimal Control)

In optimal control settings, such as the Merton problem, consumption is derived endogenously by maximizing expected utility:

$$\max_{C(t),S(t)} \mathbb{E}\left[\int_0^T e^{-\rho t} U(C(t))\, dt + e^{-\rho T} U(W(T))\right]$$

subject to the wealth dynamics:

$$dW(t) = [W(t)r(t) + S(t)(\mu - r(t)) - C(t)]\, dt + \sigma S(t) dZ(t)$$

Under CRRA utility $U(C) = \frac{C^{1-\phi}}{1-\phi}$, the optimal consumption often takes the form:

$$C(t) = \frac{1}{\psi} W(t)$$

where ψ is a constant derived from preference parameters such as ρ and ϕ, as well as market coefficients.

Habit Formation (Advanced)

To model lifestyle anchoring and relative utility:

$$C(t) = \max(W(t) - H(t), 0), \qquad dH(t) = \lambda\,(C(t) - H(t))\,dt$$

Use case: Captures the psychological cost of consuming below past accustomed levels. This introduces memory into preferences and increases the HJB's dimensionality.

Summary of Modeling Choices

Model Type	**Formula**	**Notes**
Constant	$C(t) = c$	Simplest exogenous form.
Proportional to Wealth	$C(t) = \kappa W(t)$	Intuitive, scale-invariant, and tractable.
Optimal (CRRA)	$C(t) = \frac{1}{\psi} W(t)$	Derived from first-order condition in HJB.
Habit Formation	$C(t) = W(t) - H(t)$	Path-dependent, captures lifestyle inflation.
Cyclical	$C(t) = A\sin(\omega t)$	Stylized consumption behavior.

I recommend the following consumption specification:

$$\boxed{C(t) = \kappa W(t), \quad \text{for some } \kappa \in (0,1)}$$

Maintains Analytical Tractability:

- Allows value function to retain homothetic form.
- If utility is CRRA, the ansatz $J(t, r, w) = G(t, r) \cdot \frac{w^{1-\phi}}{1-\phi}$ remains valid.

Economically Realistic:

- Matches empirical patterns of proportional consumption.
- Avoids introducing new path-dependent variables like $H(t)$.

Calibration-Friendly:

- The parameter κ can be interpreted as the marginal propensity to consume.
- It can be estimated from household-level or macroeconomic data.

Wealth Process with Habit Formation

$$\begin{aligned} dW^S(t) = {} & S(t)\,(\mu\,dt + \sigma\,dZ_t) + (W(t) - S(t))r(t)\,dt - (\vartheta + \beta - d)S(t)\,dt - C(t)\,dt \\ & - \lambda\,(W(t) - H(t))\,dt \end{aligned}$$

where the habit level $H(t)$ evolves as:

$$dH(t) = \rho\,(W(t) - H(t))\,dt$$

Term Breakdown

- $S(t)\,(\mu\,dt + \sigma\,dZ_t)$: Return from wealth invested in the risky asset.
- $(W(t) - S(t))r(t)\,dt$: Return from the wealth invested in the risk-free asset.
- $-(\vartheta + \beta - d)S(t)\,dt$: Adjustment for transaction costs, taxes, and dividends.
- $-C(t)\,dt$: Explicit consumption outflow.
- $-\lambda\,(W(t) - H(t))\,dt$: **Habit penalty** — penalizes wealth above the long-run average level.
- $dH(t) = \rho(W(t) - H(t))dt$: Evolution of the **habit level**, modeled as an exponentially weighted average of wealth.
- $\lambda > 0$: Strength of habit-formation penalty.
- $\rho > 0$: Rate at which the habit level adjusts to wealth.

Key Features of Wealth SDE with Habit Formation

Feature	**Description**
Habit Formation	Modeled via an exponentially weighted average $H(t)$ of past wealth.
Consumption Adjustment	The drift penalizes excess wealth relative to habit, making utility relative.
Behavioral Realism	Captures lifestyle inflation and the psychological cost of reducing consumption.
History Dependence	$H(t)$ introduces path dependence into the optimal control problem.
Dual Benchmarking	Agent compares both to the market (via $S(t)$) and their own past (via $H(t)$).

Chong's Proposed Equations

TODO: Try to propose one/two system of SDEs to build the Model, needed to consider the future risk rate and regret.

Section 3: Model Setup – SDE Choice Rationale

Below is my "thinking-out-loud" for each SDE to include in Section 3, with both the mathematical form and economic motivation.

3.1 Short-Rate Process (Vasicek / Ornstein–Uhlenbeck)

$$dr_t = \kappa\,(\theta - r_t)\,dt \;+\; \vartheta\,dZ_t$$

- **Why this SDE?**
  - Captures realistic mean reversion toward long-run level θ.
  - Gaussian transitions preserve analytical tractability in the HJB.
  - Directly extends Ihedioha & Osu's OU form (their Eq. (31a)).
- **Key Features:**

- Linear drift toward θ.
- Constant diffusion yields normal law for r_t.

- **Parameter Meanings:**
  - $\kappa > 0$: speed of mean reversion.
  - θ: long-run average short rate.
  - $\vartheta > 0$: volatility of rate shocks.
  - Z_t: Brownian motion driving r_t.

- **Associated Asset Price:**

$$\frac{dG_t}{G_t} = r_t\,dt, \quad G_0 = 1, \quad G_t = \exp\Bigl(\int_0^t r_s\,ds\Bigr).$$

3.2 Risky Asset Process (Geometric Brownian Motion)

$$\frac{dS_t}{S_t} = \mu_S\,dt + \sigma_S\,dW_t$$

- **Why this SDE?**
  - Standard equity model yielding log-normal returns.
  - Isolates the impact of stochastic r_t and regret.
  - Fully tractable in both HJB analysis and simulation.

- **Key Features:**
  - Constant drift μ_S and constant volatility σ_S.
  - Independent Brownian motion W_t (or specified correlation with Z_t).

- **Parameter Meanings:**
  - μ_S: expected instantaneous return of the risky asset.
  - $\sigma_S > 0$: volatility of the risky asset.
  - W_t: Brownian motion driving equity shocks.

3.3 Contribution Benchmark

$$M_t = \int_0^t c_s\,ds \quad \Longrightarrow \quad dM_t = c_t\,dt, \quad M_0 = 0.$$

- **Why this definition?**
  - Directly measures investor's cumulative net deposits.
  - Ensures regret is defined relative to true contributions.

- **Key Feature:** Deterministic flow of contributions.
- **Parameter Meaning:** c_t is the contribution (or withdrawal) rate.

3.4 Wealth Dynamics

$$dW_t = \Big[r_t(1-\alpha_t)W_t + \mu_S\,\alpha_t W_t - c_t\Big]\,dt + \sigma_S\,\alpha_t W_t\,dW_t, \quad W_0 > 0.$$

- **How we derive it:** Self-financing portfolio with contributions.
- **Why this SDE?**
  - Allocates $\alpha_t W_t$ to the risky asset and $(1-\alpha_t)W_t$ to the risk-free asset.
  - Subtracts $c_t\,dt$ for cash flows into the benchmark.
- **Term Meanings:**
  - $\alpha_t \in [0,1]$: fraction of wealth in S_t.
  - $r_t(1-\alpha_t)W_t$: drift from risk-free holding.
  - $\mu_S\,\alpha_t W_t$: expected drift from risky holding.
  - $-c_t\,dt$: outflow to benchmark M_t.
  - $\sigma_S\,\alpha_t W_t\,dW_t$: diffusion from equity exposure.

1. OU (Vasicek) vs. CIR vs. Cyclical Drift

Feature	**OU (Vasicek)**	**CIR**	**Asymmetric Cycles**
Mean reversion	Yes, linear drift toward θ	Yes, linear drift toward θ	No (external cycles)
Volatility form	Constant volatility ϑ	State-dependent $\sqrt{r_t}$	Constant σ
Non-negativity	Can go negative	Stays ≥ 0 under Feller condition	Can go negative unless constrained
Tractability	Closed-form bond & normal density	Closed-form bond (with Bessel functions)	No closed-form bonds, needs numeric
Economic realism	Captures mean reversion, but allows negative rates	More realistic non-negative rates, volatility proportional to level	Captures episodic spikes, but ad hoc

To Be Decided

Why I Picked OU/Vasicek Initially

1. **W**e want to preserve the ability to apply a homothetic HARA ansatz and (ideally) recover closed-form expressions for the multiplier $h(t, r)$. The CIR extra state-dependent volatility would force us into purely numerical PDEs and obscure the clean link between rate randomness and regret behavior.
2. **Analytical Simplicity:** OU's linear drift and constant diffusion makes the HJB PDE mild—only adding simple second-derivative terms in r. We get Gaussian transition densities and can sometimes solve the joint (t, r) PDE in closed form (or via expansions).
3. **Alignment with I&O:** Their original linear $r_t = \alpha + \sigma t$ is already a deterministic process. Extending to OU keeps the same linear mean-reversion structure but adds randomness in the cleanest possible way.

When CIR Might Be Better

- **Non-negativity:** CIR ensures $r_t \geq 0$, which is appealing if you reject the possibility of negative rates outright.
- **State-dependent risk:** Volatility that grows with $\sqrt{r_t}$ captures the empirically observed fact that high-rate regimes often exhibit more turbulence.

Downside: The HJB PDE gains terms like

$$\tfrac{1}{2}\,\sigma^2\, r\, J_{rr},$$

making the joint solution in (t, r) more complex (Bessel functions in bond prices, for example).

Asymmetric Cycles: A Different Animal

- **Main:Why we avoid cyclical drift:** it breaks the homothetic reduction and forces all integrals and PDEs into heavy numerics. Our goal is to get clear comparative-statics on how rate uncertainty amplifies regret aversion—something the OU model yields transparently.
- **Pros:** Can mimic prolonged low-rate periods punctuated by sharp rate hikes—realistic for post-crisis central-bank behavior.
- **Cons:** It's an ad-hoc drift function that breaks mean reversion and destroys almost all analytic tractability. We'd be forced into full numerical PDE or Monte Carlo only.

2. Risky Asset: GBM vs. Stochastic Drift/Vol

I stuck with the classical GBM

$$\frac{dS_t}{S_t} = \mu_S\,dt + \sigma_S\,dW_t$$

to isolate rate effects.

Criterion	Stochastic μ_t, σ_t	Constant GBM (μ_S, σ_S)
State dimension	Adds at least two new SDEs (μ_t, σ_t)	Keeps state space at (W_t, M_t, r_t)
HJB dimension	5-dimensional PDE (time + W, M, r, μ, σ)	3-dimensional PDE, still difficult but manageable
Interpretability	Hard to disentangle rate vs. performance risks	Isolates the impact of stochastic rates and regret alone

James's proposal to also let μ_t and σ_t follow their own SDEs is powerful, and improve realism. but:

- Adds two extra state dimensions (μ_t, σ_t), blowing up the HJB from 3 to 5 dimensions.
- **To Be Decided:** By keeping equity dynamics simple, every shift in the optimal share can be traced back to rate risk or regret aversion. However, I believe introducing stochastic drift/vol blends multiple dynamic frictions at once is better suited to the rest of paper.

3. To Be Discussed

1. **Research Focus:** If our main novelty is on *interest-rate risk & regret*, it makes sense to keep equity dynamics simple (GBM) and choose the OU model for r_t.
2. **Tractability vs. Realism:** OU gives us a tractable extension with mean reversion and randomness, CIR adds realism at the cost of analytic complexity, and cyclical drifts force us into pure numerics.
3. **Paper Length & Audience:** A closed-form or semi-analytic solution with OU will be more elegant and easier to present; CIR could be a secondary robustness check.

Personal Idea, To Be Discussed

Personal Recommendation: Use **Vasicek (OU)** for the main model—highlight its mean reversion, ease of embedding into the HJB, and direct lineage from I&O's linear-rate case. Then, if time permits, include a **CIR robustness section** showing how non-negativity and state-dependent vol shift the optimal strategy, but relegate that to an appendix or numerical supplement.

Personal Idea: Introducing stochastic drift and volatility for the equity process (letting both μ_t and σ_t follow their own SDEs) means you're simultaneously modeling four new sources of randomness—stock returns, stock volatility, interest rates, and regret—each interacting with the others. That dramatically increases the state–space (from three dimensions (W_t, M_t, r_t) to five or more), turns the HJB into a high-dimensional PDE with very few prospects for closed-form reduction, and forces you into heavy numeric machinery just to see basic patterns.

By contrast, our primary research question is: **How does stochastic interest-rate risk, together with a contribution-based regret benchmark, reshape the optimal HARA allocation?** To answer that crisply, we keep the equity side as simple as in Merton and I&O—classical GBM—so that **every change** in the optimal policy can be traced back to either the mean-reverting r_t or the regret term $W_t - M_t$. This maintains analytical tractability (we can still use the homothetic ansatz to collapse the PDE) and yields clear closed-form or semi-analytic expressions for the optimal share. In short, a simple GBM + OU backbone lets our innovation shine through, rather than being buried under a morass of interacting stochastic processes.

Alternative SDE System

Below are the other proposed SDE systems for our model with a descriptive description, including motivation, term definitions, key features, and advantages for each SDE, followed by an overall justification for why this system will work.

1. Short-Rate Process: Cox–Ingersoll–Ross (CIR)

$$dr_t = \kappa_r(\theta_r - r_t)\,dt + \sigma_r\sqrt{r_t}\,dZ_t^r.$$

- **Motivation:**
  - Motivated by James's proposal
  - Ensures nonnegative nominal rates, matching real bond markets.
  - Mean-reversion toward a central-bank target θ_r.
  - State-dependent volatility: calm when rates are low, turbulent when high.
- **Term/Parameter Definitions:**
  - r_t: instantaneous short-term interest rate at time t
  - dr_t: change in the short-term interest rate.
  - $\theta_r > 0$: long-run average level of the short rate.
  - Z_t^r: standard Brownian motion driving for rate shocks.
  - $\kappa_r(\theta_r - r_t)\,dt$: deterministic drift pulling r_t toward θ_r.
    - $\kappa_r > 0$: speed of mean reversion to θ_r (how quickly r_t is pulled back toward θ_r).
    - $(\theta_r - r_t)$: distance from long-run mean.
    - dt: infinitesimal time increment.
  - $\sigma_r\sqrt{r_t}\,dZ_t^r$: stochastic shock to r_t.
    - $\sigma_r > 0$: volatility coefficient—scales the size of random shocks.
    - σ_r: scales volatility of shocks.
    - $\sqrt{r_t}$: state-dependent volatility—larger when r_t is high.
    - dZ_t^r: increment of standard Brownian motion.
- **Key Features:**
  - **Nonnegativity:** If
    $$2\kappa_r\,\theta_r \;\geq\; \sigma_r^2 \quad \text{(the Feller condition)},$$
    then r_t stays strictly positive.
  - **State-Dependent Volatility:** The diffusion term $\sigma_r\sqrt{r_t}$ shrinks when rates are low, matching empirical calm in low-rate regimes.
  - **Analytic Bond Pricing:** Under CIR, zero-coupon bond prices have known closed-form expressions involving confluent hypergeometric functions.
- **Advantage:** By ensuring $r_t \geq 0$ and giving volatility proportional to $\sqrt{r_t}$, CIR captures two realistic phenomena—no negative nominal rates and lower turbulence when rates are near zero—while still admitting semi-analytic solutions.

2. Variance Process: Square-Root (Heston-Style)

$$dv_t = \kappa_v(\theta_v - v_t)\,dt + \sigma_v\sqrt{v_t}\,dZ_t^v.$$

- **Motivation:**
  - To model volatility clustering—periods of high market turbulence followed by calm—and maintain nonnegative instantaneous variance.
- **Term/Parameter Definitions:**

· v_t: instantaneous variance of the risky asset return.
· dv_t: change in instantaneous variance of the risky asset.
· $\kappa_v > 0$: speed of mean reversion for variance.
· $\theta_v > 0$: long-run average variance.
· $\sigma_v > 0$: volatility of volatility, scaling random shocks' amplitude.
· Z_t^v: Brownian motion (possibly correlated with W_t^S).
· $\kappa_v(\theta_v - v_t)\,dt$: mean-reverting drift of variance.
· $\sigma_v\sqrt{v_t}\,dZ_t^v$: stochastic volatility shocks.
· $\sqrt{v_t}$: ensures variance stays nonnegative.
· dZ_t^v: Brownian increment.

– **Key Features:**
  * **Nonnegativity:** Same Feller-type condition

$$2\,\kappa_v\,\theta_v \;\geq\; \sigma_v^2$$

    keeps $v_t \geq 0$.
  * **Volatility Clustering:** High v_t begets more randomness in returns, which then mean-reverts, capturing clustering.
  * **Flexibility:** Allows skewed and heavy-tailed return distributions when plugged into a Heston-style price process.

– **Advantage:** Provides realistic stochastic volatility with well-known calibration procedures. This process gives a far more realistic model of equity returns—volatility is itself random and mean-reverting—while retaining tractability for simulation and, in some cases, semi-analytic option-pricing formulas.

3. Risky Asset Price: Heston-Style

$$\frac{dS_t}{S_t} = \mu_S\,dt + \sqrt{v_t}\,dW_t^S.$$

– **Motivation:**
  * To reflect that returns' volatility changes over time (volatility-of-volatility), not just the price level.

– **Term/Parameter Definitions:**
  * S_t: risky asset price.
  * dS_t/S_t: instantaneous return on the risky asset.
  * μ_S: expected return/ constant or slowly-varying risk premium.
  * $\mu_S\,dt$: expected drift component.
  * v_t: instantaneous variance.
  * W_t^S: Brownian motion for returns. Correlated with Z_t^v.
  * $\sqrt{v_t}\,dW_t^S$: random return fluctuation.
    · $\sqrt{v_t}$: time-varying volatility level.
    · dW_t^S: Brownian motion increment for returns.

– **Key Features:**
  * Volatility clustering via v_t./ Instantaneous volatility $\sqrt{v_t}$ captures regime shifts in market risk.
  * Flexibility to model skew and kurtosis./ Correlation between Z_t^v and W_t^s can generate leverage effects.

– **Advantage:** Integrates stochastic variance into portfolio choice, enriching risk dynamics. Reproduce empirically observed return skewness and kurtosis, improving calibration to real markets and better capturing tail-risk.

4. **Contribution Benchmark: Feedback Rule**

$$dM_t = \left[c_0 + \eta\,(W_t - M_t)\right] dt.$$

- **Motivation:**
  * To endogenize savings behavior: investors contribute more when they're ahead and less when they're behind.
- **Term/Parameter Definitions:**
  * M_t: cumulative net contributions.
  * $c_0 \geq 0$: base contribution rate.
  * η: sensitivity of contributions to wealth–benchmark gap.
  * W_t: current Wealth
  * dM_t: change in cumulative contributions.
  * $c_0\,dt$: baseline contribution flow.
  * $\eta\,(W_t - M_t)\,dt$: performance-driven adjustment.
    · $(W_t - M_t)$: current surplus (if positive) or shortfall (if negative).
- **Key Features:**
  * When $M_t > W_t$,contributions rise; if behind, contributions fall.
  * Captures effects in saving, linking regret to saving rates.
- **Advantage:** Captures behavioral saving patterns and enriches regret dynamics.

5. **Wealth Dynamics**

$$dW_t = \Big[r_t(1-\alpha_t)W_t + \mu_S\,\alpha_t W_t - \big(c_0 + \eta(W_t - M_t)\big)\Big]dt + \sqrt{v_t}\,\alpha_t W_t\,dW_t^S.$$

- **Motivation:** Combine all sources of return and saving into a self-financing framework.
- **Term/Parameter Definitions:**
  * $\alpha_t \in [0,1]$: fraction of wealth invested in S_t.
  * $r_t(1-\alpha_t)W_t$: risk-free drift.
  * $r_t(1-\alpha_t)W_t\,dt$: return from risk-free portion.
  * $\mu_S\,\alpha_t W_t$: equity expected drift.
  * $\mu_S\,\alpha_t W_t\,dt$: expected return from risky portion.
  * $-(c_0 + \eta\,(W_t - M_t))\,dt$: net contributions flow out of wealth.
  * $\sqrt{v_t}\,\alpha_t W_t\,dW_t^S$: stochastic equity return.
- **Key Features:**
  * **Multi-Source Risk:** Wealth is influenced by rate risk, equity risk, and volatility-of-volatility.
  * **Endogenous Saving:** Contributions shift with performance.
  * **Behavioral Feedbacks:** Falling behind the benchmark reduces contributions, potentially exacerbating regret.
- **Advantage:** Although high-dimensional, this SDE captures rich interactions between interest-rate risk, equity risk, saving behavior, and regret. I believe It could provide a solid foundation work and formulation for how these multiple frictions shape optimal dynamic portfolio and saving decisions. However, we might need requiring more advanced numerical methods(e.g. high-dimensional PDEs or Monte Carlo).

Dynamic Performance-Based Contribution Benchmark

Define the benchmark as follows:

$$dM_t = \left[c_0 + \eta(W_t - M_t)^+ - \xi(M_t - W_t)^+\right] dt,$$

where:

- $x^+ = \max(x, 0)$.
- $c_0 \geq 0$: baseline or minimum contribution rate (ensures steady accumulation).
- $\eta \geq 0$: sensitivity parameter when wealth exceeds the benchmark.
- $\xi \geq 0$: sensitivity parameter when wealth falls below the benchmark.

Overall Justification

This system will work for our research because:

1. Well-Posed, Arbitrage-Free Market

- **Standard Foundations:** Each process (CIR for r_t, square-root for v_t, Heston for S_t) is a classical, well-studied SDE with unique strong solutions and no arbitrage.
- **Self-Financing Wealth:** Our wealth equation follows the self-financing condition—changes come only from asset returns and contributions—so there are no hidden cash "jumps" or violations of budget constraints.

2. Captures Key Economic Facts

- **Non-negative Rates & Volatility:** CIR enforces $r_t \geq 0$ and $\sqrt{v_t} \geq 0$, matching real markets (no negative nominal rates, no negative variance).
- **Mean Reversion:** Both interest rates and volatility revert to long-run averages (θ_r, θ_v), reflecting central-bank policy and volatility clustering.

3. Markovian Structure & Control Framework

- **Finite-Dimensional Markov State:** Our state vector (W_t, M_t, r_t, v_t) is Markov and continuous, so the dynamic-programming principle applies.
- **HJB Applicability:** Even though the HJB becomes higher-dimensional, it remains a well-posed second-order PDE on this state space. We can solve it numerically (finite-difference, sparse grid) or approximate via deep-learning or perturbation methods (**TO BE DECIDED**).

4. Flexibility & Calibration

- **Parameter Interpretability:** Every parameter—$\kappa_r, \theta_r, \sigma_r$, $\kappa_v, \theta_v, \sigma_v$, c_0, η, μ_S—has clear economic meaning and can be calibrated to market data (bond yields, option-implied vol, savings rates).
- **Rich Comparative Statics:** By varying, say, η we see how more or less responsive saving alters regret dynamics; by varying σ_r or σ_v we isolate rate-risk vs. equity-risk impacts.

5. Numerical Tractability

- **Simulation-Friendly:** All four SDEs can be discretized with well-understood schemes (full-truncation Euler for CIR/Heston, standard Euler for wealth and M_t).
- **Controlled Complexity:** Yes, the state space grows, but modern PDE and Monte-Carlo techniques (e.g. regression-based least-squares) handle four or five dimensions easily for many applied problems.